

Farah Mohamed

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EDUCATION

Université de Montréal

Computer Science (B.Sc.)

Montréal, QC

2022 – 2025

PROJECTS

Collaborative Whiteboard | *Next.js, React.js, Typescript, HTML, Tailwind CSS*

[Github](#) | [Live Demo](#)

- Development of a Full Stack real-time collaborative whiteboard application allowing multiple users to draw and interact simultaneously.
- Implementation of secure authentication with Clerk and real-time synchronization with Liveblocks.
- Use of Convex as a reactive database and Next.js for a modern and high-performance web architecture.

AI Inbox Firewall | *Python, Django, React, Solace Agent Mesh, PostgreSQL, Redis, Docker, MQTT*

[Github](#)

- Developed an intelligent email inbox featuring a multi-agent architecture utilizing Solace Agent Mesh.
- Implemented 6 specialized AI agents: spam filtering, priority sorting, summarization, task extraction, tone analysis, and URL scanning (VirusTotal API).
- Orchestrated the full-stack environment using Docker Compose to containerize Django REST API, React, Solace PubSub+, PostgreSQL, and Redis.

BorderMate | *Next.js, React.js, Typescript, HTML, Tailwind CSS, REST API*

[Github](#) | [Live Demo](#)

- Automated multilingual communication application with real-time processing.
- Resilient architecture with multiple APIs for high availability.
- Collaboration using GitHub and automated deployment on GitHub Pages.

Automated Testing | *Java, Git, Maven*

[GitHub Link](#)

- Developed as part of a Software Quality course (IFT3913).
- Implemented a CI/CD pipeline using GitHub Actions to automate test execution and ensure code integrity.

Compiler Host Language Integration | *Scheme, BASIC, x86*

[Github](#)

- Four-month project conducted for a Programming Languages and Compilation course (IFT3065).
- Utilized Scheme and x86 assembly to implement a parser, lexer, and target code generator.
- Collaborated via GitHub in a team repository, resulting in the final integration of the host language.

EXPERIENCE

Security Analyst

July 2024 – June 2025

Canada Revenue Agency (CRA)

Ottawa, ON

- Analyze security risks of software and created accounts to approve or deny requests for additional permissions.
- Collaborate and communicate with all individuals within the CRA (59,045 employees).
- Escalate complex cases to a senior developer or request additional information to resolve issues.

Tutor

October 2023 – June 2025

MonsieurMath

Montréal, QC

- Personalized guidance for more than 20 students, leading to academic success with an average score above 80%.
- Optimization of remote learning through the collaborative use of Teams and Miro for the visualization of complex concepts.
- Design of iterative learning plans based on the specific objectives of each student.
- Analysis and management of pedagogical data via Excel to automate progress tracking.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, TypeScript, Javascript Haskell, C, C++, Scheme, Swift, Gambit, BASIC

Frameworks: Next.js, React, NumPy, Pytorch, Node.js, Three.js

Developer Tools: Git, VS Code, Visual Studio, Docker, Microsoft 365 suite, Figma, Miro

Languages: French (Native), English (Fluent), Spanish (Conversational)